SINGLE-PHASE · LOW-VOLTAGE BATTERY · 3 – 6 KW

OLI-1H Series
Hybrid Inverter

One compact unit for solar generation, 48 V battery storage and seamless backup — built for Indian homes and small businesses that need reliable power through outages and lower bills with time-of-use optimisation.

OLI-1H3L | OLI-1H3.6L | OLI-1H4L | OLI-1H4.6L | OLI-1H5L | OLI-1H6L | OLI-1H5L-2



200 %
PV oversizing (DC/AC) for higher yield

<10 ms
grid-to-backup switch-over

97.3 %
peak efficiency

Why OLI-1H

Up to 200 % PV input
Oversize the array on 2 MPPTs with 18 A per string — ideal for high-current modules.

Up to 200 % grid input
Fast grid charging and pass-through for heavy loads up to 10 kVA.

48 V battery flexibility
Lithium or lead-acid, 40 – 60 V, up to 135 A charge / discharge; CAN-linked to Olitec OLB packs.

Parallel up to 6 units
Scale from 3 kW to 36 kW single-phase as your needs grow.

6-period time-of-use
Charge cheap, discharge at peak; peak-shaving with grid-constraint awareness.

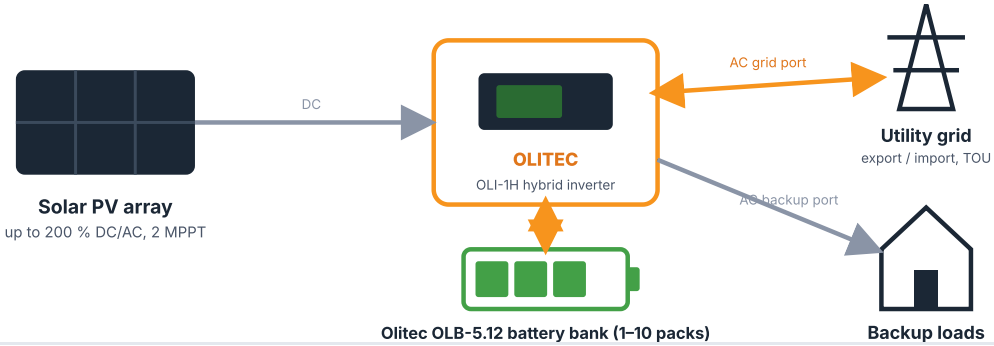
App monitoring & VPP ready
Remote monitoring, firmware updates and virtual-power-plant access.

Shadow scan
Global MPPT scan keeps production optimal on partially shaded roofs.

Complete protection
AFCI (optional), Type II SPD, insulation and leakage detection, 8-level battery protection.

IP66 · fan-less
Natural convection, < 35 dB, –20 to +60 °C — indoor or outdoor installation.

System overview



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Single-Phase Hybrid Inverter 3 – 6 kW

Low-voltage battery

Model	OLI-1H3L	OLI-1H3.6L	OLI-1H4L	OLI-1H4.6L	OLI-1H5L	OLI-1H6L	OLI-1H5L-2
DC Input (PV Side)							
Max. input power [W]	4800	5760	6400	7360	8000	9600	8000
Max. input voltage [V]	500						
Rated input voltage [V]	370						
MPPT voltage range [V]	125 – 425						
Start-up voltage [V]	125						
Max. input current per MPPT [A]	18.0 / 18.0						
Max. short-circuit current per MPPT [A]	27.0 / 27.0						
No. of MPPTs / strings per MPPT	2 / 1+1						
Battery Connection							
Battery type	Lithium-ion / Lead-acid						
Max. charge / discharge power [W]	3000 / 3000	3600 / 3600	4000 / 4000	4600 / 4600	5000 / 5000	6000 / 6000	5000 / 5000
Battery voltage range [V]	40 – 60						
Max. charge / discharge current [A]	70	90	100	110	120	135	120
AC Output (Grid Port)							
Rated output power [W]	3000	3600	4000	4600	5000	6000	5000
Max. output apparent power [VA]	3300	4000	4400	5000	5500	6600	5000
Max. input apparent power from grid [VA]	6000	7200	8000	9200	10000	10000	10000
Rated grid voltage / frequency	220 / 230 V, 50 / 60 Hz						
Rated output current [A]	13.6 / 13.0	16.0 / 16.0	18.2 / 17.4	20.9 / 20.0	22.7 / 21.7	27.3 / 26.1	22.7 / 21.7
Max. output current [A]	15.0	16.0	20.0	22.7	25.0	30.0	22.7
Max. input current from grid [A]	27.3	32.7	36.4	42	43	43	43
Power factor	~1 (adjustable 0.8 leading to 0.8 lagging)						
Current THD [%]	< 3 @ rated power						
AC Output (Backup Port)							
Rated power [W]	3000	3600	4000	4600	5000	6000	5000
Max. output apparent power [VA]	3300	4000	4400	5000	5500	6600	5000
Rated output voltage / frequency	220 / 230 V, 50 / 60 Hz						
Rated output current [A]	13.6 / 13.0	16.4 / 15.7	18.2 / 17.4	20.9 / 20.0	22.7 / 21.7	27.3 / 26.1	22.7 / 21.7
Max. output current [A]	15.0	18.2	20.0	22.7	25.0	30.0	22.7
Max. output peak power [W] / duration	6000 (<10 s)	7200 (<10 s)	8000 (<10 s)	9200 (<10 s)	10000 (<10 s)	10000 (<10 s)	10000 (<10 s)
Grid-tied to stand-alone switching time [ms]	< 10						
Output voltage THD [%]	< 3 @ linear load						
Efficiency							
Max. efficiency [%]	97.0	97.0	97.0	97.3	97.3	97.3	97.3
Euro efficiency [%]	96.0	96.0	96.0	96.2	96.2	96.2	96.2
Protection							
Safety protection features	LVRT, HVRT, anti-islanding, AC output over / under-voltage, AC short-circuit, AC over-current, leakage current, Type II DC / AC surge protection, PV insulation resistance detection, PV reverse polarity, PV input over-voltage, earthing fault protection, DC arc-fault detection (optional)						
General Data							
Dimensions (W × H × D) [mm]	228 × 470 × 340						
Weight [kg]	19						
Operating temperature range [°C]	–20 to +60 (derating above +45)						
Relative humidity / max. altitude	0 – 100 % (non-condensing) / 2000 m						
Ingress protection	IP66						
Cooling	Natural convection						
User interface	LCD, LED, mobile app						
Communication with BMS	CAN						
Typical noise emission [dB]	< 35						
Standards & Warranty							
Compliance	IEC 62109-1 & -2, IEC 61000-6-1, IEC 61000-6-3, IEC/EN 62920, IEC 60068, IEC 61683, IEC 63027, IEC 60730, EN 303645, IEC 61727, IEC 62116, EN 50549-1 / EN 50549-10, IS 16221 / IS 16169 / IS 17980						
Standard warranty [years]	5 (extended warranty available)						

OLI-1H5L-2: 5 kW variant without 10 % AC-side overload capacity (max. output apparent power 5000 VA). Rated output currents shown at 220 V / 230 V. Warranty terms per Olitec Solar warranty policy. All specifications subject to change without prior notice.



THREE-PHASE · HIGH-VOLTAGE BATTERY · 15 – 30 KW

OLI-3H Series
Hybrid Inverter

Commercial-grade three-phase hybrid with up to 60 kW of PV input, dual high-voltage battery ports and full backup output — for offices, schools, clinics, retail and light industry.

OLI-3H15H | OLI-3H20H | OLI-3H25H | OLI-3H29.9H | OLI-3H30H

60 kW	33 kW	97.8 %
max. PV input (OLI-3H30H)	battery charge / discharge power	peak efficiency

Why OLI-3H

Up to 3 MPPTs, 36 A each
Six PV strings on 25 – 30 kW models; 160 – 850 V MPPT window; 1000 V max.

160 – 750 V battery
Dual battery inputs (50 A + 50 A) on 25 kW and above; CAN-based BMS integration.

100 % unbalanced output
Full rated power on any single phase in backup mode; 160 % peak for 10 s.

Parallel up to 6 units
Up to 180 kW of hybrid capacity with AC-coupling and diesel-generator integration.

Peak shaving & TOU
Dynamic control against tariff schedules and grid-constraint limits.

Grid-code compliant
Anti-islanding, LVRT / HVRT, residual current and insulation monitoring, Type II SPD.

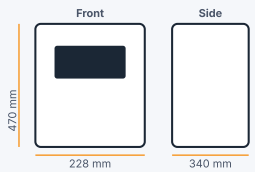
Typical applications

Commercial & industrial self-consumption
Offices, hospitals, schools, retail and small factories with 3-phase connections and daytime loads.

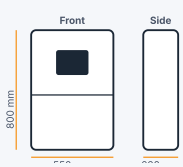
Peak shaving & demand management
Cap contracted demand and shift consumption away from peak-tariff windows.

Critical backup
Seamless three-phase backup for servers, lifts, pumps and medical equipment; DG integration for extended outages.

Dimensions



OLI-1H Series
228 × 470 × 340 mm · 19 kg · IP66
Wall-mounted, natural convection



OLI-3H Series
550 × 800 × 230 mm · 46.5 – 50 kg · IP66
Wall-mounted, intelligent fan cooling

Three-Phase Hybrid Inverter 15 – 30 kW

High-voltage battery

Model	OLI-3H15H	OLI-3H20H	OLI-3H25H	OLI-3H29.9H	OLI-3H30H
DC Input (PV Side)					
Max. input power [W]	30000	40000	50000	59800	60000
Max. input voltage [V]			1000		
Rated input voltage [V]			650		
MPPT voltage range [V]			160 – 850		
Start-up voltage [V]			180		
Max. input current per MPPT [A]	36 / 36	36 / 36	36 / 36 / 36	36 / 36 / 36	36 / 36 / 36
Max. short-circuit current per MPPT [A]	45 / 45	45 / 45	45 / 45 / 45	45 / 45 / 45	45 / 45 / 45
No. of MPPTs / strings per MPPT	2 / 2+2	2 / 2+2	3 / 2+2+2	3 / 2+2+2	3 / 2+2+2
Battery Connection					
Battery type			Lithium-ion (high-voltage)		
Max. charge / discharge power [W]	16500	22000	27500	30000	33000
Battery voltage range [V]			160 – 750		
Max. charge / discharge current [A]	50	50	50 + 50	50 + 50	50 + 50
Number of battery inputs	1	1	2	2	2
AC Output (Grid Port)					
Rated output power [W]	15000	20000	25000	29900	30000
Max. output apparent power [VA]	16500	22000	27500	29900	33000
Max. input apparent power from grid [VA]	30000	40000	50000	60000	60000
Rated grid voltage / frequency			3L / N / PE, 220 / 380, 230 / 400 V, 50 / 60 Hz		
Rated output current [A]	22.8 / 21.8	30.4 / 29	37.9 / 36.3	45.4 / 43.4	45.4 / 43.5
Max. output current [A]	25 / 24	33.4 / 31.9	41.7 / 39.9	45.4 / 43.4	50 / 47.9
Max. input current from grid [A]	45.5 / 43.3	60.6 / 57.7	76.0 / 72.2	90 / 86.6	90 / 86.6
Power factor			~1 (adjustable 0.8 leading to 0.8 lagging)		
Current THD [%]			< 3 @ rated power		
AC Output (Backup Port)					
Rated power [W]	15000	20000	25000	29900	30000
Max. output apparent power [VA]	16500	22000	27500	29900	33000
Rated output voltage / frequency			3L / N / PE, 220 / 380, 230 / 400 V, 50 / 60 Hz		
Rated output current [A]	22.8 / 21.8	30.4 / 29	37.9 / 36.3	45.4 / 43.4	45.4 / 43.5
Max. output current [A]	25 / 24	33.4 / 31.9	41.7 / 39.9	45.4 / 43.4	50 / 47.9
Max. output peak power [W] / duration	30000 (<10 s)	40000 (<10 s)	40000 (<10 s)	48000 (<10 s)	48000 (<10 s)
Grid-tied to stand-alone switching time [ms]			< 10		
Output voltage THD [%]			< 3 @ linear load		
Efficiency					
Max. efficiency [%]			97.8		
Euro efficiency [%]	96.7	97.2	97.2	97.2	97.2
Protection					
Safety protection features	Anti-islanding, DC reverse polarity, insulation monitoring, residual current monitoring, Type II DC / AC surge protection, DC arc-fault detection (optional), three-phase unbalanced output				
General Data					
Dimensions (W × H × D) [mm]			550 × 800 × 230		
Weight [kg]	46.5	46.5	50	50	50
Operating temperature range [°C]			–25 to +60		
Relative humidity / max. altitude			0 – 100 % / 3000 m		
Ingress protection			IP66		
Cooling			Intelligent fan cooling		
User interface			LCD, LED, mobile app		
Communication with BMS			CAN		
Typical noise emission [dB]	< 55	< 55	< 60	< 60	< 60
Standards & Warranty					
Compliance	IEC/EN 62109-1 & -2, IEC/EN 61000-6-1 / -2 / -3 / -4, IEC/EN 62920, IEC 60068-2-1 / -2 / -14 / -27 / -30 / -64, IEC 61683, EN 50530, EN 50549-1 & -10, IEC 61727, IEC 62116, NRS 097-2-1				
Standard warranty [years]			5 (extended warranty available)		

Rated output currents shown at 220 V / 230 V phase voltage. OLI-3H29.9H is rated to 29.9 kW for connections limited below 30 kW. Warranty terms per Olitec Solar warranty policy. All specifications subject to change without prior notice.